

Technical Datasheet — t8_r3 V13 porousanode (VBF-T8R3-DSH-01)

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Virtual-design datasheet; values from parameter set and simulation outputs (no physical cell exists).

	Value	Source
Capacity	5.0 Ah nominal (parameter set); 5.0728 Ah simulated 1C DFN	Chen2020; r4_V13_porousanode_1c_dfn.json
Voltage range / window	window 2.5–4.2 V; discharge midpoint 3.957 V	Chen2020 cut-offs; energy:midpoint_voltage_v
Energy	18.568 Wh	energy:energy_wh ($\int V \cdot I \, dt$, 1C DFN)
Energy density	497.54 Wh/kg (962.7 Wh/L)	contract formula: energy ÷ calc-energy stack mass (electrolyte excluded)
Continuous discharge	5C verified: 5.005 Ah, retention 98.66% vs 1C	r4_V13_porousanode_5c_dfn.json / derived
Temperature capability	4C/45 °C charge: T_max 326.53 K (limit 333.15)	4c_safety:T_max_K
Temperature range	verified 298.15 K (discharge) and 318.15 K ambient (charge); 5C discharge peak cell temperature 314.0 K	protocol definitions; 5c_dfn:T_max_K
Notes	Not simulated (task has no cycle-life criterion; no aging protocol run) — must not be fabricated	log.json! stage4 note
Termination	no plating (anode min +0.0177 V); overcharge→thermal runaway ODE: triggered=False	4c_safety / run-tr
Volume and mass	65 × 1580 × 187.8 μm; stack 37.32 g (electrolyte excluded) + 6.79 g electrolyte fill	parameter set + energy:mass_kg
Internal resistance (10% point)	1.27 mΩ	energy:dcr_ohm